

Benjamin Garcia

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EDUCATION

University of California, Los Angeles

Bachelor of Science, Computer Science

Los Angeles, CA

Expected Graduation: Jun. 2027

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, C, C++, SQL, JavaScript, HTML/CSS

Frameworks & Tools: Next.js, React, Tailwind CSS, FastAPI, Prisma, Node.js, Redis, PostgreSQL, Supabase, Firestore, JIRA

Cloud & DevOps: AWS Lambda, Palantir Foundry, Docker, Vercel, Git, CI/CD, Linux/Unix, GitLab, Kubernetes

EXPERIENCE

Interface Undergraduate Researcher

Dec. 2025 – Present

UCLA Human-Computer Interaction Research Lab

Los Angeles, CA

- Extending **GraPPI**'s knowledge-graph-based **GraphRAG** framework by designing **human-in-the-loop** interfaces in React, TypeScript, and Next.js that expose interaction graphs, edge-level explanations, and path-level reasoning outputs to support protein-protein interaction (PPI) exploration.
- Building interactive workflows that surface intermediate edge explanations, retrieved STRING annotations, relevance scores, and re-ranking logic from the **retrieve-divide-solve** pipeline, enabling domain experts to fact-check and refine AI-recommended PPI pathways.
- Integrating multi-agent LLM services via Python and WebSockets to support real-time hypothesis iteration, structured path evaluation, and transparent explanation rendering for **CHI/IUI/NAACL** research on explainable biomedical AI systems.

Software Engineer Intern

Mar. 2025 – Sep. 2025

Todd Agriscience

Remote, Part-time

- Delivered AI-driven crop insight dashboards for **5–10** clients by architecting full-stack systems in Next.js, Tailwind CSS, and Framer Motion, integrated with Palantir Foundry and AWS Lambda.
- Developed dynamic data pipelines visualizing **20+** agricultural data streams (soil, irrigation, weather, yield) to enable real-time farm management decisions.

Research and Development Engineer Intern

Jul. 2025 – Aug. 2025

Bonterra

Remote

- Prototyped an agentic AI system for nonprofit event analytics using tool-calling frameworks and Model Context Protocols (MCPs), modeling attendance, retention, and fundraising across datasets; benchmarked **5+** agent frameworks.
- Developed and tested a multi-tool orchestration layer enabling autonomous agent collaboration, improved retrieval latency by **22%**, and validated tool-selection policies for scaling across nonprofit datasets.

Software Engineer Intern

May 2025 – Jun. 2025

TensorStax

Remote

- Engineered real-time agentic UIs with Next.js, Redux, and WebSockets, supporting **100+** concurrent users with sub-**100ms** latency while collaborating with a **10–15**-engineer team across frontend, ML, and infrastructure.
- Built secure credential-submission workflows integrated with HashiCorp Vault and a Python + Redis backend, unifying authentication across **50+** enterprise data sources (AWS Glue, Snowflake, MongoDB, Postgres, Airflow, dbt, Spark).

PROJECTS

Git Proof | *Next.js, TypeScript, PostgreSQL, Prisma, Gemini 2.5, GitHub GraphQL API*

Dec. 2025 – Present

- Architected a full-stack developer portfolio platform transforming GitHub activity into recruiter-ready insights, processing **100+** repositories per user via a type-safe pipeline integrating the GitHub GraphQL API, PostgreSQL, and Prisma ORM.
- Designed a **3-factor** weighted impact scoring algorithm (0–50 scale) combining logarithmic popularity scaling, recency decay across **5** time tiers, and project maturity signals to fairly rank diverse portfolios with user-level weighted aggregation.
- Built an AI-powered insights and content system using Google Gemini 2.5 Flash, generating context-aware project descriptions, professional bios, and personalized portfolio feedback by analyzing repository metadata, documentation quality, and contribution patterns.
- Implemented security- and reliability-first infrastructure, including read-only GitHub OAuth scopes, database-stored tokens, complete account deletion with OAuth grant revocation, and a hybrid caching strategy with TTL-based freshness checks to handle API rate limits while maintaining data integrity.

Poly Clubs | *Next.js, TypeScript, Supabase, PostgreSQL, Tailwind CSS*

Dec. 2025 – Present

- Built a full-stack review platform for **493** student organizations using Next.js 15 and Supabase, supporting anonymous reviews with browser fingerprinting and database-level rate limiting to prevent abuse.
- Designed a serverless moderation and data access layer using PostgreSQL RPC functions and Row Level Security, reducing client-server round trips by **60%** and achieving **10–100x** faster queries via indexed aggregation.
- Architected a security-first validation model with multi-layer constraints (client, server, database), integrated FingerprintJS for cross-session enforcement, and implemented dynamic OpenGraph metadata across **493** statically generated pages.